Base58

An easy-to-share set of characters

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Base58 Character Set

```
0 1 2 3 4 5 6 7 8 9
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
```

Base58 is a **user-friendly set of characters** you can use to represent big numbers in a shorter format.

Satoshi came up with this character set in the first release of Bitcoin ↗. It is used for encoding legacy `addresses`, WIF private keys, and extended keys.

Terminology

What does "base58" mean?

The "*base*" refers to the *number of characters* you use to represent a value.

Base	Characters
2 (binary)	`01`
10 (decimal)	`0123456789`
16 (hexadecimal)	`0123456789abcdef`
58	`123456789ABCDEFGH JKLMN PQRSTUVWXYZabcdefghijk mnopqrstuvwxyz`

In everyday life, we are used to working with **base10** numbers (using the ten digits `0123456789`).

But if you're a computer, it's easy enough to use *extra characters* to represent values:

```
base10(9999) = 9999
base16(9999) = 270f
base58(9999) = 3yQ
```

All of these "numbers" have the same *value* – they just use **different character sets** (i.e. *bases*) to represent them.

Tip: The more characters you have in your *base*, the fewer characters you need to use to represent big numbers.

Benefits

Why do we use Base58 in Bitcoin?

```
// Why base-58 instead of standard base-64 encoding?
// - Don't want 0OIl characters that look the same in some fonts and
//      could be used to create visually identical looking account numbers.
// - A string with non-alphanumeric characters is not as easily accepted as an account number.
```

```
// - E-mail usually won't line-break if there's no punctuation to break at.
// - Doubleclicking selects the whole number as one word if it's all alphanumeric.
```

– Satoshi Nakamoto, *Bitcoin v0.1 (`base58.h`) ↗*

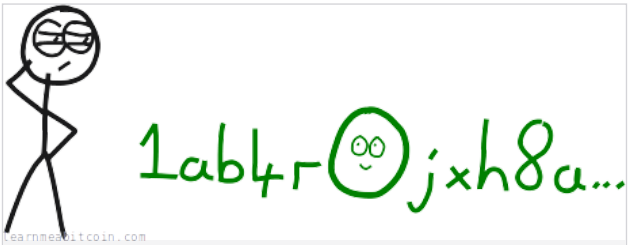
Why Base*58*? Why these specific 58 characters?

Because these are the characters you are left with when you use all the characters in the alphanumeric alphabet (62 in total), but **remove the easily mistakable characters** `0`, `O`, `l`, and `I`.

```
alphanumeric = 0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
base58       =  123456789ABCDEFGH JKLMN PQRSTUVWXYZabcdefghijk mnopqrstuvwxyz
```

So base58 has two main benefits:

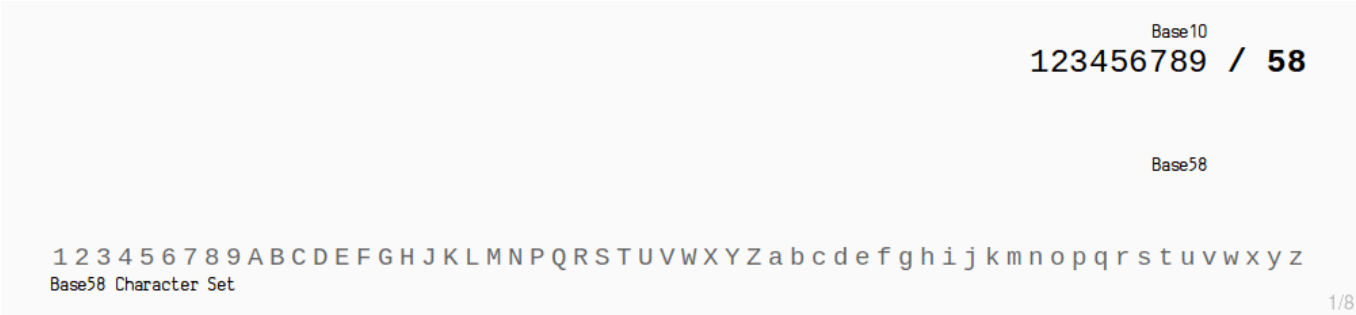
1. **It gives you a bigger set of characters to work with.** This means you can represent larger numbers using fewer characters.
2. **It leaves out awkward characters**. This helps to prevent mistakes when transcribing.



A pesky O/0

Encode

Convert an integer to Base58



To convert an integer (base10) to base58, you use the *modulo*[1 ↘] function.

Basically, you keep dividing your number by 58, taking the **remainder** at each step of the way to get the **index** of the next base58 character, and you stop when there are no remainders left.

For example:

```
base10 = 123456789

123456789 % 58 = 19 <- remainder
  2128565 % 58 = 23 <- remainder
    36699 % 58 = 43 <- remainder
      632 % 58 = 52 <- remainder
       10 % 58 = 10 <- remainder
```

```
base58 = [10][52][43][23][19]
base58 = BukQL
```

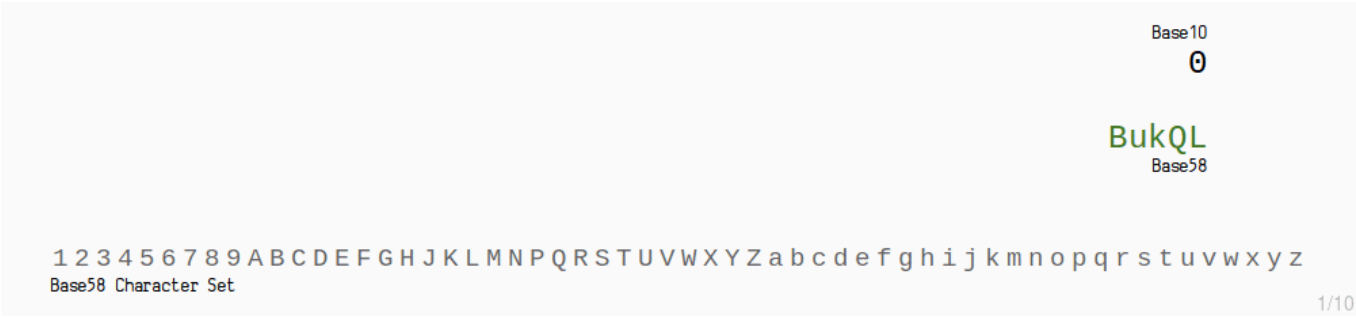
Code [hide]

Ruby

```ruby
# A simple function that converts an _integer_ to base58:

def int_to_base58(i)

  characters = %w[
  1 2 3 4 5 6 7 8 9
  A B C D E F G H J K L M N P Q R S T U V W X Y Z
  a b c d e f g h i j k m n o p q r s t u v w x y z
  ]

  # create an empty string (in preparation to hold the new characters)
  buffer = ''

  # keep finding the remainder until our starting number hits zero
  while i > 0
    # find the remainder after dividing by 58 (% = modulo)
    remainder = i % 58

    # add the base58 character to the start of the string
    buffer = characters[remainder] + buffer

    # divide our integer by 58, and repeat...
    i = i / 58
  end

  return buffer
end

puts int_to_base58(123456789) #=> BukQL
```

Decode

Convert Base58 to an integer



To convert a base58 value into base10 (integer), working from right to left, you take the **index** for each Base58 character and multiply it by **increasing powers of 58**, then add all the values together.

For example:

```
base58 = BukQL

L = 19 * 58^0 = 19
Q = 23 * 58^1 = 1334
k = 43 * 58^2 = 144652
u = 52 * 58^3 = 10145824
B = 10 * 58^4 = 113164960

base10 = 19 + 1334 + 144652 + 10145824 + 113164960
base10 = 123456789
```

Code [hide]

Ruby

```ruby
def base58_to_int(base58)

  characters = %w[
      1 2 3 4 5 6 7 8 9
    A B C D E F G H   J K L M N   P Q R S T U V W X Y Z
    a b c d e f g h i j k   m n o p q r s t u v w x y z
    ]

  # create an integer to hold the result
  total = 0

  # reverse the base58 string so we can read characters from right to left
  base58 = base58.reverse

  # run through each character, including the index, so we know how many characters we've read
  base58.each_char.with_index do |char, i|

    # get the index number for this character
    char_i = characters.index(char)

    # work out how many 58s this character represents (increment the power for each character)
    value  = char_i * (58**i)

    # add to total
    total = total + value
  end

  return total
end

puts base58_to_int("BukQL") #=> 123456789
```

Tip: This mathematical process is the same when converting from one base to another. You can see a similar example when converting from hexadecimal to decimal (base16 to base10).

Base58 in Bitcoin

How is Base58 used in Bitcoin?

Base58 is used in Bitcoin when you want to convert *commonly used data* into an easier-to-share format. For example:

- **`Addresses`**
  - A `public key` is the "public" counterpart to a `private key`, and you use them when you want to send bitcoins to someone. However, public keys are quite lengthy, so we convert them to addresses instead (which makes use of base58 in the final step of the conversion).
- **WIF Private Keys**
  - A `private key` is like a "password" that you use to unlock bitcoins for spending, and sometimes you may want to import one into a wallet. For this occasion there is such a thing as a WIF Private Key, which is basically a private key converted to base58.
- **Extended Keys**
  - If you're using an HD Wallet, you may come into contact with `extended private keys` and `extended public keys`. These are also converted to base58 to shorten them and make them easier to share.

> **Tip:** Base58 was only used for encoding `addresses` at first, but was utilized for encoding WIF Private Keys and extended keys when they were introduced later on.

Leading Zeros

```
00123456789abcdef
↓
1C3CPq7c8PY
```

In Bitcoin, we convert every *zero byte* (`0x00`) at the start of a hexadecimal value to a `1` in base58.

Putting zeros at the start of a number does not increase its size (e.g. `0x12` is the same as `0x0012`), so when we convert to base58, any additional zeros at the start do not affect the result.

Therefore, to ensure that leading zeros have an influence on the result, the bitcoin base58 encoding includes a *manual step* to convert all leading `0x00`'s to `1`'s.

> **Note: 0x:** A `0x` prefix indicates a hexadecimal value. Hexadecimal values will sometimes only contain the numbers 0-9 and could therefore be confused as being decimal values, so the prefix helps us to distinguish between them. This prefix is discarded before being used in calculation.

> **Note: Byte:** A byte of data can hold a value of between `0-255`, and can be represented by *two* hexadecimal characters. For example, `0xff` is one byte of data and represents the value `255` in decimal.

Prefixes

In Bitcoin, different **prefixes** are added to data before converting to base58 to influence the **leading character** of the result, and this leading character helps us to identify what each base58 string represents.

These are the most common prefixes used in Bitcoin:

Mainnet | Testnet

Hex Prefix	Base58 Leading Character	Represents	Example
`00`	`1`	Address (P2PKH)	`1AKDDsfTh8uY4X3ppy1m7jw1fVMBSMkzjP`

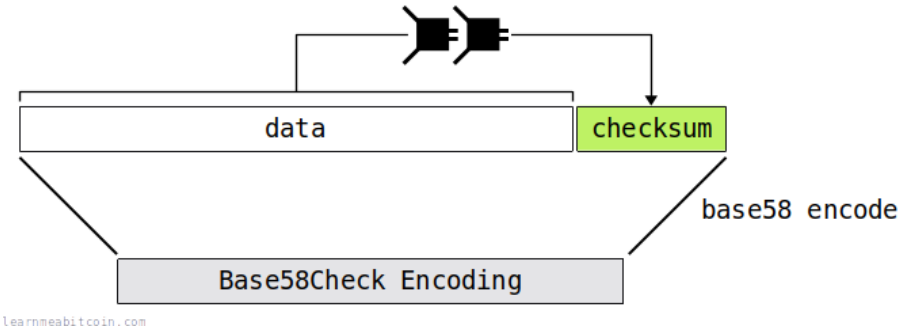
Hex Prefix	Base58 Leading Character	Represents	Example
05	3	Address (P2SH)	34nSkinWC9rDDJiUY438qQN1JHmGqBHGW7
80	K, L, or 5	WIF Private Key	L4mee2GrpBSckB9SgC9WhHxvtEgKUvgvTiyYcGu38mr9CGKBGp93
0488ADE4	xprv	Extended Private Key	xprv9tuogRdb5YTgcL3P8Waj7REqDuQx4sXcodQaWTtEVFEp6yRKh1CjrWfXChnhgHeLDuXxo2auDZegMiVMGGxwxcrb2PmiGyCngLxvLeGsZRq
0488B21E	xpub	Extended Public Key	xpub67uA5wAUuv1ypp7rEY7jUZBZmwFSULFUArLBJrHr3amnymkUEYWzQJz13zLacZv33sSuxKVmerpZeFExapBNt8HpAqtTtWqDQRAgyqSKUHu

https://en.bitcoin.it/wiki/List_of_address_prefixes ↗

Tip: WIF Private Keys use the same hex prefix, but produce different leading characters. This is because in some situations we *append* an 01 byte to the private key before converting to base58, and this extra byte affects the leading character.

Note: Extended Keys contain extra metadata alongside the original public and private keys, which is why their base58 strings are noticeably longer.

Base58Check



Base58Check encoding is short for **adding a checksum** to some data before encoding it to base58.

Here are some common examples of Bitcoin data that use Base58Check encoding:

- Address (Base58) [+]
- WIF [+]
- Address (Extended Key) [+]

You sometimes see this "Base58Check" term pop up every now and then when reading about base58 encoding, so I thought I'd cover it here.

Code

These code snippets perform the complete **base58 conversion used in Bitcoin**.

They convert to and from hexadecimal, because that's the most common format we start with when converting to base58.

Ruby PHP

Ruby

```
module Base58

  @chars = %w[
      1 2 3 4 5 6 7 8 9
    A B C D E F G H   J K L M N   P Q R S T U V W X Y Z
    a b c d e f g h i j k   m n o p q r s t u v w x y z
]
  @base = @chars.length

  def self.encode(hex)
    i = hex.to_i(16)
    buffer = String.new

    while i > 0
      remainder = i % @base
      i = i / @base
      buffer = @chars[remainder] + buffer
    end

    # add '1's to the start based on number of leading bytes of zeros
    leading_zero_bytes = (hex.match(/^([0]+)/) ? $1 : '').size / 2

    ("1"*leading_zero_bytes) + buffer
  end

  def self.decode(base58)
    total = 0 # integer to hold conversion to decimal

    # run through each character
    base58.reverse.each_char.with_index do |char, i|
      char_i = @chars.index(char) # get the index number for this character
      value  = (58**i) * char_i   # work out how many 58s this character represents
      total = total + value     # add to total
    end

    # convert this integer to hex
    hex = total.to_s(16)

    # add leading 00s for every leading 1
    leading_1s = (base58.match(/^([1]+)/) ? $1 : '').size

    ("00"*leading_1s) + hex
  end

end

puts Base58.encode('0093ce48570b55c42c2af816aeaba06cfee1224faebb6127fe') #=> 1EUXSxuUVy2PC5enGXR1a3yxbEjNWMHuem
puts Base58.decode('1EUXSxuUVy2PC5enGXR1a3yxbEjNWMHuem') #=> 0093ce48570b55c42c2af816aeaba06cfee1224faebb6127fe
```

Notes

Modulo (%)

The modulo ↗ (%) operator returns the ***remainder* of a division**:

```
7 % 6 = 1
7 % 5 = 2
7 % 4 = 3
7 % 3 = 1
```

It's like a sister of the divide (`/`) operator.

Resources

- src/base58.cpp ↗
- github.com - base58.rb ↗
- bitcoin.it/wiki/Base58Check_encoding ↗
- darklaunch.com - Base58 Encoder/Decoder ↗

Further Reading

- purplemath.com - Number Bases ↗

Thanks

- TheSeven ↗ - Chatted on IRC. Helped me to understand how Base58 is implemented in Bitcoin.